

**Abstract**—Click-Through Rate (CTR) is an important measure of how effectively search results attract user clicks. Pages that receive many search impressions but relatively few clicks represent valuable opportunities for content optimisation. This study investigates whether machine learning can predict CTR using anonymised search performance data from the FlyRank ML Internship Warehouse. Daily search metrics were aggregated to the page level using DuckDB, and regression models were trained to estimate CTR from historical search performance features. Experimental results show that simple regression models can identify patterns associated with CTR while also highlighting the limitations of the available features. The proposed workflow provides a repeatable and evidence-based approach for prioritising pages that may benefit from search optimisation.

**Index Terms**—Learning Outcomes: Machine Learning, Search Intelligence, Click-Through Rate, Regression, Search Optimization, Data Analytics

## I. INTRODUCTION

Modern websites generate large volumes of search performance data every day. As the number of pages grows, manually identifying content that requires optimisation becomes increasingly difficult. Machine learning offers an opportunity to analyse historical search behaviour and assist content teams in prioritising pages that are most likely to benefit from improvement. This project uses anonymised search performance data from the FlyRank ML Internship Warehouse as a case study. The focus is on pages that receive substantial search visibility but relatively low Click-Through Rates (CTR), since improving these pages can increase organic traffic without creating new content. A regression-based machine learning pipeline was developed to estimate CTR using historical search metrics. The resulting predictions provide a practical ranking of optimisation opportunities while demonstrating how data-driven methods can support search intelligence workflows.

## II. PROBLEM STATEMENT

This study aims to find pages that have high visibility in search results and low CTR. These pages might need to be optimized for title, meta description or content.

The research question is:

Are there any pages that show meaningful opportunities to improve CTR if analyzed by machine learning models, based on past search performance?

## III. DATASET

This study uses the FlyRank ML Internship Warehouse dataset.

The analysis is based on the `fact_content_daily_performance` table.

Using DuckDB, data from daily observations were aggregated into one row per content page.

The following features were used:

- Impressions
- Clicks
- Average Search Position
- Pageviews
- Sessions
- Scroll Events

Click-Through Rate (CTR):

$$CTR = \frac{\text{Clicks}}{\text{Impressions}} \quad (1)$$

## IV. METHODOLOGY

### A. Data Processing

The Hugging Face warehouse was queried directly and without downloading the entire dataset using DuckDB.

Statistics were computed on a daily basis and aggregated to the page level to be suitable for machine learning.

### B. Feature Engineering

The following predictor variables were chosen:

- Impressions
- Average Position
- Pageviews
- Sessions
- Scroll Events

CTR was used as the prediction target.

### C. Model Development

The data set was split into two parts: training data (80%)  
Two regression models were tested:

- Linear Regression
- Random Forest Regressor

### D. Evaluation Metrics

Performance of the models was assessed based on:

- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)
- $R^2$  Score

## V. RESULTS

The regression results are summarised in Table I. Linear Regression achieved an MAE of 0.00906 with an  $R^2$  value of 0.02475, while the Random Forest model achieved a slightly lower MAE of 0.00850 but a negative  $R^2$  score. Although Random Forest reduced the average prediction error, the negative  $R^2$  indicates that it did not generalise well to unseen data. Overall, the results suggest that the available features explain only a limited amount of the variation in CTR Table I.

TABLE I  
REGRESSION MODEL PERFORMANCE

| Model             | MAE     | MSE     | $R^2$    |
|-------------------|---------|---------|----------|
| Linear Regression | 0.00906 | 0.00118 | 0.02475  |
| Random Forest     | 0.00850 | 0.00148 | -0.21712 |

## VI. EXPLORATORY DATA ANALYSIS

Data samples of CTR and search performance metrics were analysed to get an understanding of their distribution.

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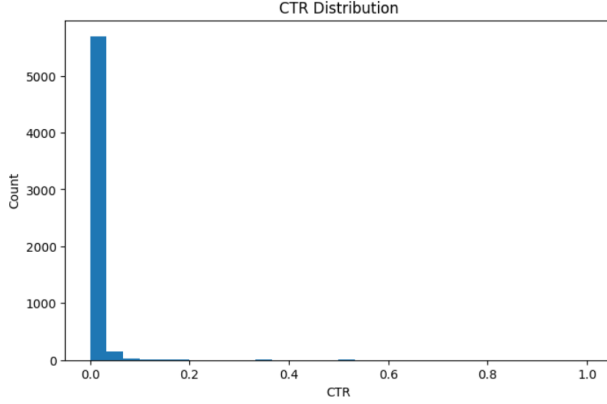


Fig. 1. Distribution of Click-Through Rate

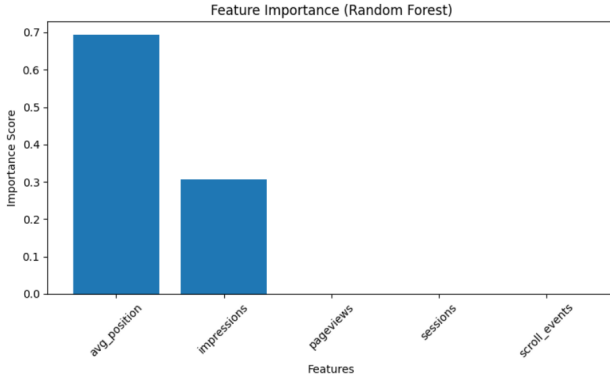


Fig. 2. Feature Importance

## VII. FEATURE IMPORTANCE

Feature analysis indicates that impressions, average search position, sessions and page engagement metrics contribute most to CTR prediction. These variables provide useful signals for identifying pages that may benefit from optimisation, although additional behavioural and content-quality features would likely improve predictive performance.

## VIII. RANKED RECOMMENDATIONS

Predicted CTR values were used to rank pages according to their optimisation potential. Pages with high search visibility but relatively low predicted CTR should be prioritised for review. Recommended actions include:

- Improve page titles and meta descriptions.
- Update content to better match user search intent.
- Improve user engagement and page experience.
- Monitor CTR after optimisation to evaluate improvements.

## IX. LIMITATIONS

This study has several limitations.

- The analysis uses historical observational data and cannot establish causal relationships.
- Only aggregated page-level metrics were included as model features.
- Content quality, textual features and backlink information were unavailable.
- The relatively low  $R^2$  values indicate that additional predictors are needed for stronger performance.

Therefore, the model should be viewed as a decision-support tool rather than a complete optimisation solution.

## X. CONCLUSION

This study demonstrates a complete machine learning workflow for predicting Click-Through Rate using search performance data from the FlyRank ML Internship Warehouse. DuckDB enabled efficient processing of large-scale search data, while regression models provided an evidence-based approach for identifying pages with optimisation potential. Although prediction accuracy was limited by the available features, the workflow illustrates how machine learning can support search intelligence by helping content teams prioritise optimisation efforts. Future work should incorporate richer behavioural and content-based features to improve predictive performance.

## ACKNOWLEDGMENT

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